1(a)(i)	(halogens can) remove electron(s) (from a reactant)	**1**
1(b)(i)	**M1** Br—Br is weaker (than C*l*—C*l*) **AND** Br atoms are larger **M2** less (significant) overlap of orbitals (in Br-Br)	**2**
1(b)(iii)	**M1** (enthalpy change / energy change) when one mole of a compound / substance is formed **M2** from its (constituent) elements in their standard state(s)	**2**
1(b)(iv)	**M1** (6 × –92) – (2 × –46) = **M2** –460 ($kJ\,mol^{-1}$)	**2**

Examiner report missing
9701_w25_qp_24 Question 1

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**
3(d)(i)	HC*l* (g)	**1**
3(d)(ii)	**M1** (reaction 1): –3115 –286 + 3292 = –109 (kJ mol^{-1}) **M2** ΔH_2 = – 53 – (–109) = (+)56 (kJ mol^{-1})	**2**

Examiner report missing
9701_w25_qp_23 Question 3

3(a)	**M1** (family of molecules / compounds / series with) same general formula **OR** (family of molecules where) molecular formula differs (only) by (units of) $CH_{2(n)}$. **M2** similar chemical properties / reactions	**2**

3(e)(i)	H_2 ((g)) **OR** hydrogen (gas)	**1**

3(e)(ii)	**M1** +215 + (–156) – (+48) = **M2** +11 kJ mol^{-1}	**2**

Examiner report missing
9701_w25_qp_22 Question 3

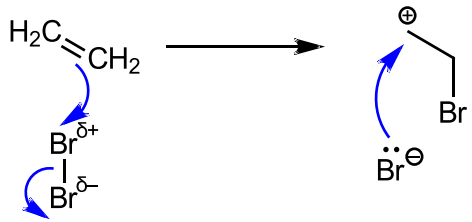
3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**
3(d)(i)	HC*l* (g)	**1**
3(d)(ii)	**M1** (reaction 1): –3115 –286 + 3292 = –109 (kJ mol^{-1}) **M2** ΔH_2 = – 53 – (–109) = (+)56 (kJ mol^{-1})	**2**

Examiner report missing
9701_w25_qp_21 Question 3

3(a)(i)	two (or more) reactants / molecules react to make (only) one product	**1**

3(c)	**method 1** – breaking all bonds in molecules $(2 \times 610) + (2 \times 436) + (10 \times 410) + (3 \times 350) – (14 \times 410) – (5 \times 350)$ • energy involved when all bonds are broken = (7242) • energy involved when all bonds are made = (7490) • energy to break bonds – energy to make bonds = (7242 – 7490) = –248 **method 2** – ignore any bonds that do not change from reactants to products $(2 \times 610) + (2 \times 436) – (4 \times 410) – (2 \times 350)$ • energy involved when all bonds are broken = (2092) • energy involved when all bonds are made = (2340) • energy to break bonds – energy to make bonds (2092 – 2340) = –248	**2**

Examiner report missing
9701_s25_qp_22 Question 3

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**

4(b)	= –37.8 (kJ mol^{-1})	**1**

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

3(a)	$H_2(g) + ½O_2(g) \rightarrow H_2O(l)$ **M1** correct equation only **M2** correct state symbols based on this equation	**2**

3(b)	$\Delta H = -(-1264) - (2 \times -46) + (2 \times -134) + (3 \times -286) = (+)230\ kJ\ mol^{-1}$ **M1** correct expression in terms of all 4 enthalpy changes of formation **AND** correct stoichiometry (as described in given equation) **M2** correct calculation based on figures given	**2**

Question 3

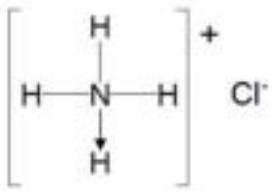
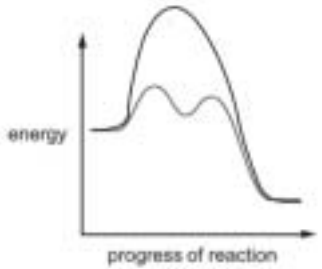
Candidates are urged to look carefully at the information provided in each item: here, this proved helpful and key to answering the ensuing physical chemistry questions.

(a) This was well answered by candidates. Standard states were required, that of water being (l).

(b) Common errors included incorrect stoichiometry or reverse parity for the answer. On the whole, the principle of Hess's Law seemed to be understood where answered.

(c)(i) There were very many different answers here; it seemed that candidates were supposing that the reaction of 'metallic' boron and non-metallic carbon must result in a giant ionic lattice.

(ii) This was well answered by candidates who correctly used the A_r values of B and C. Some confusion was evident over the method of deciding the ratio of B : C. Answers without working were not credited, nor were answers trying to reason a formula from the ratio of putative charges of ions.

4(a)	 **OR** Correct representation or label showing **M1** (ionic bond between) NH_4^+ and Cl^- **AND** 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 **M2** 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons1	**2**
4(d)	 **M1** same start energy and finishing energy **AND** single hump **M2** transition state / activation energy shown in answer is higher than both transition state energies shown in reaction profile (as the reaction is slower)	**2**

Question 4

Many individual parts of this question were answered well. Parts **(a)**, **(c)(i)**, **(e)(ii)** and **(f)(ii)** were clearly the most challenging, often as candidates were required to apply several independent parts of their knowledge of the syllabus.

(a) Candidates were not strictly required to give dot-and-cross diagrams, though these and modified Lewis diagrams often produced the clearest responses. Many answers incorrectly showed a pentavalent nitrogen atom — a kind of NH_5 species, with little understanding of the NH_4^+ Cl^- superstructure. The dative bond needed to be clearly distinct from other covalent bonds and ionic charges needed to be given.

(b)(i) There were many correct answers to this question.

(ii) Different versions of this answer were offered, but a large number were a simple addition of NH_4^+ and OH^-, which did not gain credit.

(c)(i) This was well answered by candidates whose knowledge extended to terminology associated with electron-pair donors.

(ii) This was found to be the most challenging item on the paper which required candidates to assess either the basicity or nucleophilicity of each species. Opposing answers were therefore acceptable with enough reasoning.

(d) This was generally well answered by candidates, but the single-step nature of the S_N2 mechanism needed to be clearly evidenced by one transition state; the slower rate of this reaction by a higher energy of the TS relative to either produced in S_N1. Candidates frequently accessed the former part but often then erroneously showed some kind of catalysed reaction with a lower activation energy.

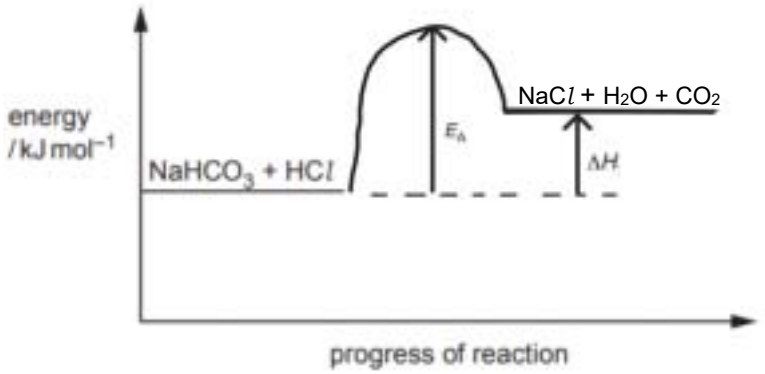
(e)(i) This question was not generally well answered, as candidates were not able to associate electron movement with the correct stage of the reaction and often were very imprecise with their positioning of curly arrows. Many answers had the arrowhead and tail reversed, with a lot of responses suggesting electron movement *from* extra H^+ species.

(ii) This question was reasonably well answered.

(iii) This question was generally well answered, although stray vowels or unspecific/incorrect numbering were observed. Candidates are reminded that the longest carbon chain is the important root of a molecule's name.

(f) (i) Candidates were able to provide a correct answer here by analogy of the 2-bromobutylium species on the left-hand side.

(ii) This was found to be a demanding question, although candidates who recognised the necessary reference to the stability of the intermediate were able to give a reasonably straightforward AO1 account for full credit.

2(a)	 *All three correct (2 marks) or two correct (1 mark):* • profile for endothermic reaction (include 'hump') • arrow representing E_A • arrow representing ΔH **OR** (+) 27.2	**2**
2(b)(i)	no more fizzing / effervescence stops	**1**
2(b)(ii)	**M1** calculate Q from = 50 × 4.18 × (26.2 – 19.6)) = 1379.(4) **M2** *Use M1 to find energy change when 1 mol Na_2CO_3 / 2 mol HCl reacts* **M1** ÷ 0.04 = 34485 **M3** *express Q as $\Delta H_{(1)}$ in kJ mol^{-1}* = – (value for **M2**) ÷ 1000 = –34.5 kJ mol^{-1} *Alternative method* **M1** *evidence of use $Q = mc\Delta T$* = 50 × 4.18 / 1000 × (26.2 – 19.6) = 1.3794 **M2** *Use M1 to find energy change when 1 mol Na_2CO_3 / 2 mol HCl reacts* **M1** ÷ 0.04 = 34.485 **M3** *express Q as $\Delta H_{(1)}$ in kJ mol^{-1}* = – (value for **M2**) *(expressed in kJ)* = –34.5 kJ mol^{-1}	**3**

2(b)(iii)	*Using answer from 2(b)(ii)* (=) $2\Delta H_2 - (\Delta H_1)$ (=) $2 \times 27.2 - (2b(ii))$ $2 \times 27.2 - (-34.50) = (+)\ 88.9\ kJ\ mol^{-1}$ *All three correct (2 marks) or two correct (1 mark):* • (27.2) – (–34.5) OR (27.2) – (2b(ii)) • 2×27.2 and $1 \times$ 2b(ii) • calculate correctly their expression Using **alternative** value of $\Delta H_1 = -38.4\ kJ\ mol^{-1}$ $\Delta H_r = (+)92.8\ kJ\ mol^{-1}$ = 2 marks	**2**

Question 2

(a) Appropriate profiles for an endothermic reaction were common. In general, representation of activation energy and enthalpy change were correct. Some answers showed very imprecise location of arrows on the diagram or showed arrows which did not point in the correct direction.

(b) (i) Good answers described an appropriate observation based on the idea that no more gas is produced when the reaction stops.

(ii) This calculation proved to be demanding. The specific heat capacity of water was not always given to the appropriate number of significant figures. The step to convert the amount of energy produced in the experiment to the amount of energy produced per mol was often confused. A significant proportion of answers included a negative sign for the enthalpy change value to reflect the exothermic nature of the reaction. Some answers were given in terms of J mol^{-1} rather than kJ mol^{-1}, as directed in the question.

(iii) The best answers showed a correct Hess' cycle in their working. Weaker responses tended to use appropriate stoichiometry in their calculation but confused the algebraic relationship between ΔH_1 and ΔH_r in the calculation. A relatively high proportion of candidates did not attempt to answer this question.

(c) Many answers deduced that **Z** was a Group 2 nitrate. Identification of calcium as a Period 4 element presented some difficulties and $Sr(NO_3)_2$ was seen on occasions. Production of the balanced equation for this thermal decomposition proved to be demanding.
The weakest responses identified the brown gas as bromine and gave equations of Group 2 bromides producing bromine.

4(a)	**M1** (enthalpy change / energy change) when one mole of a compound / substance is formed **M2** from its elements in their standard states	**2**

4(b)(i)		equation for reaction	$\Delta H_{formation}$	**2**
	Fe_2O_3	$2Fe\,(s) + \frac{3}{2}O_2\,(g) \rightarrow Fe_2O_3\,(s)$	–824.2	
	CO	$C\,(s) + \frac{1}{2}O_2\,(g) \rightarrow CO\,(g)$	–110.5	
	CO_2	$C\,(s) + O_2\,(g) \rightarrow CO_2\,(g)$	–393.5	

4(b)(ii)	$\Delta H_{reaction} = -\Delta H_{Fe2O3} - (3 \times -\Delta H_{CO}) + (3 \times -\Delta H_{CO2})$ $= -(-824.2) - (3 \times -110.5) + (3 \times -393.5)$ $= -24.8\ kJ\ mol^{-1}$ • *evidence of use of correct expression and / or any rearrangement of only these bond energy values* • *use correct stoichiometry for these bond energy values only* • *calculate their expression correctly provided the expression uses only addition / subtraction and all 3 values*	**2**

Question 4

(a) The majority of candidates recognised that the definition of enthalpy change of formation involved an enthalpy change for the formation of one mole of compound from its elements. Most answers omitted that the elements were required to be in their 'standard states'.

(b) (i) The award of full credit was rare. Many equations omitted the state symbols and/or gave correctly balanced equations for two moles of product. It was common to see Fe^{3+}, for Fe, and O for O_2.

(ii) The calculation of the enthalpy change for the reaction: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ proved challenging for many candidates. The common errors included:

- use of incorrect stoichiometry
- use of an enthalpy value for the formation of Fe and
- incorrect use of enthalpy cycles.

3(a)	strong triple bond / high triple bond enthalpy	**1**
	non-polar (molecule)	**1**

3(c)(i)	0 for both ΔH_f of N_2 and H_2	**1**
	−46 for ΔH_f of NH_3	**1**

Question 3

This question tested candidates' abilities to apply their knowledge in slightly unfamiliar contexts. In-depth knowledge and recall of nitrogen chemistry was crucial, as was the ability to draw clear, labelled, unambiguous diagrams.

(a) Different types of answer were needed to gain full credit. A strong triple bond and high E_a are in effect making the same point.

(b) (i) There were many correct answers here, with balancing being the main difficulty.

(ii) Very few answers correctly referenced both acidic species, HNO_3 and HNO_2. The latter was rarely seen.

(iii) Candidates deployed a full range of answers referring to environmental issues: specifically, this was looking at photochemical smog.

(c) (i) Candidates found it challenging to apply the definition of standard enthalpy change of formation; attempting to conceive of two numbers that would sum to –46 (or –92), or that combined with the ΔH_f would sum to zero.

(ii) Many candidates were able to give good definitions here.

(iii) Candidates did not perform well on this question. Despite '**rate**' being in bold in the question, the large majority of responses attempted to answer in terms of yield and equilibrium position. Nevertheless, some credit was available for correct and non-contradicted information about a decrease in rate. Where collision theory is referenced, candidates are reminded of the need to discuss the *frequency* (i.e., number *per unit time*) of collisions, rather than simply the number.

(d) (i) Many diagrams showed a triple bond between the nitrogen atoms or miscounted the number of non-bonding electrons.

(ii) Candidates did not show a good understanding of hybridisation as responses divided reasonably equally between those who offered sp, sp^2 and sp^3 as answers.

(iii) Some good answers were seen here, but many answers could not be awarded full credit. *Sideways* overlap of p orbitals was needed in the response, or a clear and labelled diagram showing this as part of an unambiguous representation of a pi bond.

1(a)(i)	total # 3p e^- 3 4 5	**1**
	total # unpaired e^- 3 2 1	**1**

1(e)(i)	enthalpy change when one mole of a compound / substance is formed	**1**
	from its constituent elements in their standard states	**1**

1(e)(ii)	$2(-592) = 2(-289) + (496) - 2(P{=}O)$ OR (2) $P{=}O = 2(-289) + (496) + 2(592)$	**1**
	$E_{P=O} = ½(1102) = +551$ (kJ mol^{-1})	**1**

Question 1

(a) (i) Most candidates were able to complete the table to show the number of electrons in the 3p subshells of P, S and Cl, as 3, 4 and 5 respectively.

The total number of unpaired electrons in these elements proved challenging for some candidates, Common incorrect answers were 1, 0 and 1 respectively.

(ii) The equation to represent the first ionisation of P was well answered. A common error was to omit the state symbols of (g) throughout.

(ii) The question required candidates to choose one of three possible values, given in the question, to represent the first ionisation energy of S and to explain their choice.

Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ mol^{-1}, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

Many candidates chose a value between P and Cl and supported their choice with a incorrect statement that the 'nuclear attraction to the outer electrons' increased across the period.

(b) (i) The full electronic configuration of the P^{3-} ion was well answered as: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.

(ii) The decrease in ionic radius from P^{3-} to S^{2-} and Cl^- was well answered. Many correct explanations included mention of an increased attraction of the outer electron to the nucleus resulting from an increased nuclear charge with each ionic species having the same shielding.

(c) Observations were required for three tests carried out on separate samples of NaCl(aq).

When Br_2(aq) is added to a sample no reaction occurs, and the colourless solution of NaCl(aq) turns orange (because of the added Br_2).

Most answers stated that the observation from the addition of concentrated H_2SO_4 to NaCl(aq) is misty/steamy fumes, not recognising that the NaCl is in aqueous solution (and not a solid). The solution remains colourless as there is no visible reaction.

A white precipitate is formed when $AgNO_3$(aq) is added to NaCl(aq).

(d) (i) This question provided information that the chemical behaviour of $POCl_3$ is similar to that of PCl_5; $POCl_3$ has a low melting and boiling points and reacts vigorously with water suggesting that the structure and bonding of $POCl_3$ is simple covalent.

A small number of candidates gave a good explanation for this information and appreciated that the '*simple*' structure of $POCl_3$ was related to weak intermolecular forces of attractions and that the '*covalent*' nature of the compound is linked to the *hydrolysis* reaction with water.

(ii) The equation for the reaction of $POCl_3$ with water was generally quite well answered. Errors included an incorrect formula for H_3PO_4 and/or incorrect ratio between the water molecules and HCl product.

(iii) Most candidates were awarded full marks for the dot-and-cross diagram. A common error was the use of a dative bond between the P and O atoms.

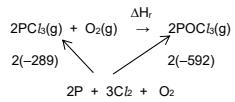
(e) (i) Most candidates were awarded the first mark for the definition of the enthalpy change of formation, for stating 'enthalpy change when 1 mole of a compound is formed'. However, very few correctly stated the second requirement that it is 'from its elements in their standard states'.

Many answers mentioned standard conditions which was not a requirement.

(ii) The calculation of the bond energy of P=O (in PCl_3), using ΔH_f and bond energy (O_2) data, proved challenging for most candidates.

Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

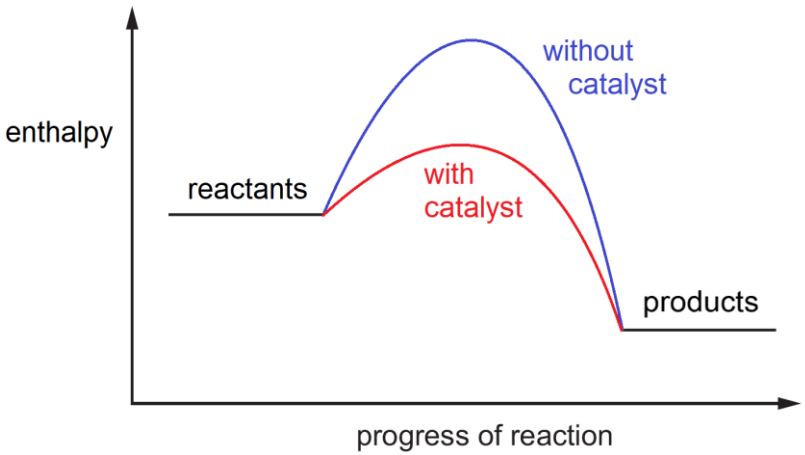
A correct energy cycle is:



ΔH_r = 2(–592) – 2(–289) = -606 kJmol^{-1}

bonds broken	bonds formed
~~6 P-Cl~~	~~6 P-Cl~~
1 O=O (496)	2 P=O

496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**
3(b)	 uncatalysed hump	**1**
	catalysed hump lower than uncatalysed	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

3(b)(i)	enthalpy change when one mole of a compound / substance is formed	**1**
	from its constituent elements in their standard states	**1**

3(b)(ii)	$\Delta H_f = +9 + 3(-1281) - 4(-972) = (+)\underline{54}$ (kJ mol^{-1}) (1)	**2**
	Any two from (1): • use correct stoichiometry for all three values • use correct expression regardless of stoichiometry for all three values • calculated correctly	

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

1(a)(i)	total # 3p e^- 3 4 5	**1**
	total # unpaired e^- 3 2 1	**1**

1(e)(i)	enthalpy change when one mole of a compound / substance is formed	**1**
	from its constituent elements in their standard states	**1**

1(e)(ii)	$2(-592) = 2(-289) + (496) - 2(P{=}O)$ OR (2) $P{=}O = 2(-289) + (496) + 2(592)$	**1**
	$E_{P=O} = ½(1102) = +551$ (kJ mol^{-1})	**1**

Question 1

(a) (i) Most candidates were able to complete the table to show the number of electrons in the 3p subshells of P, S and Cl, as 3, 4 and 5 respectively.

The total number of unpaired electrons in these elements proved challenging for some candidates, Common incorrect answers were 1, 0 and 1 respectively.

(ii) The equation to represent the first ionisation of P was well answered. A common error was to omit the state symbols of (g) throughout.

(ii) The question required candidates to choose one of three possible values, given in the question, to represent the first ionisation energy of S and to explain their choice.

Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ mol^{-1}, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

Many candidates chose a value between P and Cl and supported their choice with a incorrect statement that the 'nuclear attraction to the outer electrons' increased across the period.

(b) (i) The full electronic configuration of the P^{3-} ion was well answered as: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.

(ii) The decrease in ionic radius from P^{3-} to S^{2-} and Cl^- was well answered. Many correct explanations included mention of an increased attraction of the outer electron to the nucleus resulting from an increased nuclear charge with each ionic species having the same shielding.

(c) Observations were required for three tests carried out on separate samples of NaCl(aq).

When Br_2(aq) is added to a sample no reaction occurs, and the colourless solution of NaCl(aq) turns orange (because of the added Br_2).

Most answers stated that the observation from the addition of concentrated H_2SO_4 to NaCl(aq) is misty/steamy fumes, not recognising that the NaCl is in aqueous solution (and not a solid). The solution remains colourless as there is no visible reaction.

A white precipitate is formed when $AgNO_3$(aq) is added to NaCl(aq).

(d) (i) This question provided information that the chemical behaviour of $POCl_3$ is similar to that of PCl_5; $POCl_3$ has a low melting and boiling points and reacts vigorously with water suggesting that the structure and bonding of $POCl_3$ is simple covalent.

A small number of candidates gave a good explanation for this information and appreciated that the '*simple*' structure of $POCl_3$ was related to weak intermolecular forces of attractions and that the '*covalent*' nature of the compound is linked to the *hydrolysis* reaction with water.

(ii) The equation for the reaction of $POCl_3$ with water was generally quite well answered. Errors included an incorrect formula for H_3PO_4 and/or incorrect ratio between the water molecules and HCl product.

(iii) Most candidates were awarded full marks for the dot-and-cross diagram. A common error was the use of a dative bond between the P and O atoms.

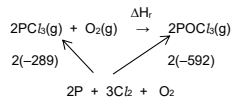
(e) (i) Most candidates were awarded the first mark for the definition of the enthalpy change of formation, for stating 'enthalpy change when 1 mole of a compound is formed'. However, very few correctly stated the second requirement that it is 'from its elements in their standard states'.

Many answers mentioned standard conditions which was not a requirement.

(ii) The calculation of the bond energy of P=O (in PCl_3), using ΔH_f and bond energy (O_2) data, proved challenging for most candidates.

Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

A correct energy cycle is:

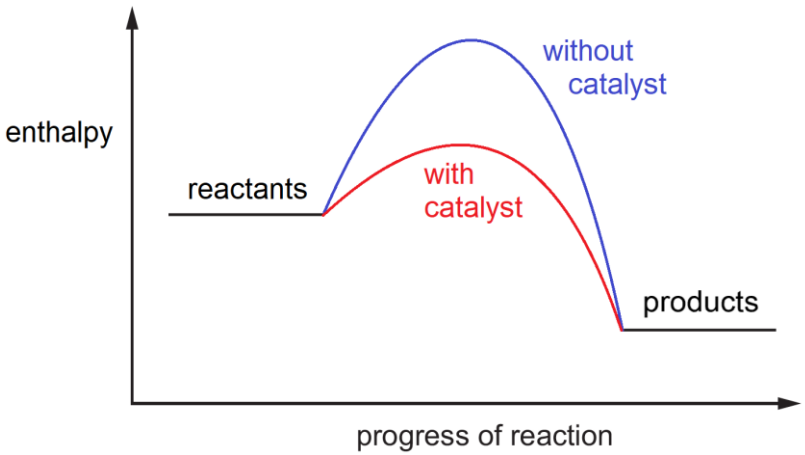


ΔH_r = 2(–592) – 2(–289) = -606 kJmol^{-1}

bonds broken	bonds formed
~~6 P-Cl~~	~~6 P-Cl~~
1 O=O (496)	2 P=O

496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(b)	 uncatalysed hump	**1**
	catalysed hump lower than uncatalysed	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

3(a)	**M1** (energy change) when 1 mole of water is formed **M2** from an (aqueous) acid and an alkali / aqueous base	**2**
3(b)(i)	0.05 mol (H_2SO_4(aq))	**1**
3(b)(ii)	**M1** *correct value for Q (using 0.1 mol H_2O / **OR** 2× (b)(i)) (and Q in Joules)* **M2** *calculate* ΔT *(using* their *Q* **AND** 110*)* $Q/mc=\Delta T$ = (Q) ÷ (4.18 × 110) = 12.42 (°C) **M3** use ΔT and initial temperature to find temperature of final solution 21.4 + M2 = 33.8 (°C) [M3 ΔT + 21.4 = T_{max}]	**3**
3(c)(i)	$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$	**1**
3(c)(ii)	(additional) energy is involved / required to fully ionise / dissociate (weak / ethanoic acid) **OR** (additional) energy is involved / required to donate/ release H^+ ions	**1**

Question 3

(a) Many candidates incorrectly stated, 'enthalpy change when 1 mole of acid reacted with 1 mole of base'.

(b) (i) Many candidates calculated the initial amount of H_2SO_4 instead of calculating the amount of acid that was neutralised by the NaOH.

(ii) Very few candidates deduced that Q = 5710 J and the mass of the solution was often incorrectly given as either 1.00 g or a number derived from calculating the masses of NaOH and H_2SO_4 in the reaction.

(c) (i) Some good answers were seen.

(ii) Several candidates knew that CH_3COOH was not fully dissociated, as it is a weak acid, compared to the strong acid HC*l*, which resulted in a less negative value for the $\Delta H_{neut.}$ Very few candidates knew that the difference in values of the respective ΔH_{neut} values was due to the additional energy required to fully dissociate the weak acid such that less energy is released after neutralisation.

4(a)(i)	$2KCl + H_2SO_4 \rightarrow K_2SO_4 + 2HCl$ **OR** $KCl + H_2SO_4 \rightarrow KHSO_4 + HCl$	**1**

4(c)(i)	new mixture will have a lower / smaller conc / amount of H_2 and I_2	**1**

4(c)(ii)	endothermic **AND** decrease in temperature causes position of equilibrium to shift in exothermic direction	**1**

Question 4

(a) (i) Some good answers were seen.

(ii) Very few candidates knew that hydrogen iodide is not prepared in the same way as hydrogen chloride because the hydrogen iodide and concentrated H_2SO_4 would undergo a redox reaction to form a variety of products to include SO_2, H_2S and I_2.

(b) (i) A small number of candidates recognised that when a sample of colourless HI(g) is placed in a sealed container and allowed to reach equilibrium the colour changes to purple, attributed to the formation of I_2(g).

(ii) This question was well answered by many candidates.

(iii) The calculation of the concentration of iodine in an equilibrium mixture was generally not well answered. Candidates who deduced that [HI] is (1.70 ÷ 2 =) 0.85 mol in the equilibrium mixture tended to gain some credit. Very few used the expression $\sqrt{(0.0217 \times 0.85)}$ to calculate the $[I_2]$ as 0.125 mol dm^{-3}.

(c) (i) Some candidates were able to describe the difference in composition of the equilibrium mixture at the two temperatures.

(ii) Some candidates deduced that the reaction is endothermic in the forward direction leading to a lower concentration of H_2 and I_2 in the equilibrium mixture.

4(a)	Method 1 • identification of number and type of bonds when all are broken 2O-H + 4C-H + C=C **OR** (2 × 460) + (4 × 410) + 610 • Identification of number and type of bonds when all are made O-H + 5C-H + C-C + C-O **OR** (460 + (5 × 410) + 350 + 360) • calculation of energy to break bonds – energy to make bonds 3170 – 3220 = – 50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks. **OR** Method 2 • identification of number and type of relevant bonds broken (including the new C-H only) 2O-H + C=C **OR** (2 × 460) + 610 • identification of number and type of relevant bonds made (including the new C-H only) O-H + C-H + C-C + C-O **OR** (460 + 410 + 350 + 360) • calculation using energy to break bonds – energy to make bonds 1530 – 1580 = –50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks.	**2**

Question 4

(a) Stronger candidates showed clear logical working and selection of appropriate data. Common errors included mis-reading the table, using an inappropriate value for a correctly identified bond, or ignoring the C-O bond during the reaction.

(b) Reference to the effect of increasing pressure on the amount of ethanol produced at equilibrium was not always included. Many explanations were given in terms of the change producing a shift to the position of equilibrium to the right; the significance of producing a smaller amount of gas to explain this change was mentioned less frequently.

(c) (i) Many candidates were able to deduce the behaviour of H_3PO_4 as an acid or an electrophile. Some explanations were not specific to step 1 in the reaction.

(ii) Some candidates were able to identify the correct species in step 2. Explanations were often vague, for example, 'seeking electrons' or 'being attacked by a pair of electrons' rather than 'accepting a lone pair'.

(iii) A small proportion of candidates applied knowledge of the drawing of mechanism arrows to represent electron movement when bonds are made and broken in this unfamiliar context. A common error showed arrows pointing in the opposite direction to the way an electron pair moves when a bond is made or broken. Sometimes an arrow did not start from the bond being broken or from the lone pair of electrons on an atom that is used to make a specific bond.

(iv) It was generally well known how a catalyst affects a reaction. Some ambiguity was seen in answers that described the effect of catalysts as increasing the number of collisions between particles rather than increasing the frequency of successful collisions.

(v) Stronger candidates clearly described the regeneration of the catalyst. Some candidates were vague and imprecise, for example 'the catalyst did not change in the reaction'.

(vi) Some candidates showed an excellent understanding of the consequence of more alkyl groups present on the C^+ of the intermediate in terms of a larger inductive effect. Some candidates incorrectly referred to the inductive effect of propan-2-ol rather than the carbocation intermediate which was responsible for the production of propan-2-ol.

(d) The sideways overlap of two p orbitals to form a pi bond between the two carbon atoms was well known. Descriptions of the formation of the sigma bond by direct overlap of sp^2 hybrid orbitals were less common.

2(a)	**M1** (enthalpy change) when 1 mole of water is formed **M2** from an (aqueous) acid and an alkali / (aqueous) base	**2**
2(b)(i)	0.05 mol	**1**
2(b)(ii)	**M1** *correct volume and correct temperature change* **M2** *calculation to calculate Q based on M1 volume and temp and Q = mc∆T* Q = *mc∆T* = 175 × 4.18 × (27.8 – 20) = (–)5705.7 **M3** *energy change when 0.1 mol H_2O is made with appropriate unit and sign* ∆*H* = M2 / 0.1 = –57.1 kJ mol^{-1} OR –57100 J mol^{-1}	**3**
2(c)(i)	..(1)..H_2SO_4(aq) + ..(1)..$Ba(OH)_2$(aq) → …(1)..$BaSO_4$(s) + 2H_2O(l)…… **M1** correct state symbols for both products **M2** correctly balanced equation	**2**
2(c)(ii)	precipitation (of $BaSO_4$) (also) involves an energy change	**1**

Question 2

(a) Some candidates were able to give a concise and accurate definition. Overall, the recall of this definition was not well attempted. Candidates often omitted any reference to a molar scale, although 'one mole of acid reacting with one mole of base' was commonly seen as an inadequate response. A common misconception was that neutralisation requires energy.

(b) (i) Few candidates were able to incorporate the relevant stoichiometry involved into their calculations.

(ii) Very few candidates gained full credit here. A very small proportion of candidates correctly used 175 g as the correct mass, many instead quoting 1 g. Many candidates converted the temperature to Kelvin but erroneously adding 273 to 7.8. A very small proportion of candidates correctly divided the energy released by the number of moles of water formed.

(c) (i) This was well answered by most candidates. Most were able to provide a correct balanced equation. Some gave the incorrect state of barium sulfate.

(ii) Many candidates stated that $BaSO_4$ is insoluble, though not many were able to link this to an additional energy change. Several answers incorrectly referred to the solubility of $Ba(OH)_2$.

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) **one product**	**1**

3(d)(i)	increases the rate of reaction	**1**
	by providing an alternative reaction pathway of lower E_a	**1**

3(d)(iii)	–187 –(–111) –(+52)	**1**
	= –128 (kJ mol^{-1})	**1**

Question 3

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

(a) (i) Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

(ii) This was well answered by most candidates.

(iii) The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

(iv) Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

(b) (i) Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the CN^- ion.
- giving the intermediate a positive charge which was subsequently attacked by the CN^- nucleophile.
- mechanisms involving the H (of HCN) / H^+ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

(ii) The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

(iii) There were some very good answers here, although some increased the number of H atoms in the product and/or H_2 appeared as a product.

(iv) This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

(c) It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

(d) (i) Some answers referred to a change in position of the equilibrium/yield, which was not credited.

(ii) Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

(iii) Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

(iv) Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.